

TexMix

An R Package for Teaching Demographic, Migration, and
Spatial Data Analysis

Mae Hutchison

2026-07-27

Agenda

I. Introduction

II. Literature Review

III. Study Area and Data Sources

IV. Analysis / Methodology

V. Results and Discussion

VI. Conclusions and Future Study

VII. References / Bibliography

I. Introduction

What is TexMix?

- An **R** package built to support **teaching spatial demographic analysis** at UT Dallas
- Bundles curated Census/ACS data, spatial data frames, and purpose-built functions
- Designed for teaching and student's own applied research
- Goal: provide a comprehensive package for teaching courses that use **R**, with easily interpretable code that students can use as a reference when writing their own scripts

Why TexMix?

- Off-the-shelf Census workflows require repetitive boilerplate: pulling ACS variables, joining geometries, managing GEOIDs
- Migration and flow data are especially painful to reshape for mapping and vector-based analysis
- TexMix pre-packages the data and functions our students and collaborators use most often
- This release focuses on **new data coverage, new analytical functions, and performance / packaging improvements**

Release Highlights

- New 2024 ACS data for NCTCOG tracts and blockgroups, plus new factor variables and new 2024 ACS data for all Texas counties
- Expanded highway and water spatial layers and added a new Texas neighbors `sdf` for basemaps
- New Italy migration dataset and basemap files
- Three new functions: `flow2vec`, `ageStructure`, `viewPaletteSpread`; updated `mapColorRamp`
- Vectorized performance improvements, universal GEOID keys, and a new testing framework

II. Literature Review

Foundations in Spatial Demography

- Long tradition of using Census geography (tracts, block groups, counties) as the building blocks of demographic analysis
- R's spatial ecosystem has matured into the de facto standard for reproducible Census workflows
- Migration and commuting flow analysis has its own lineage: origin-destination matrices, desire-line mapping, and flow vectorization techniques used in transportation and migration research

Positioning TexMix

- TexMix builds on, rather than replaces, the `sp` / `sf` ecosystem
- Distinguishing contributions:
 - Pre-joined, analysis-ready ACS extracts for a defined study region (NCTCOG) and for Texas statewide
 - Purpose-built helpers for **flow data** (`flow2vec`) and **age-structure diagrams** (`ageStructure`) that are otherwise assembled ad hoc
 - Cartographic helper functions (`mapColorRamp`, `viewPaletteSpread`) aimed at consistent, accessible thematic mapping
- Related work: regional council data portals (NCTCOG)

- Related work: regional council data portals (NCTCOG), Census Bureau flow reports, and prior TexMix releases that established the base package structure

Gaps TexMix Addresses

- Regional (NCTCOG) data is published separately from US Census and statewide Texas data – TexMix unifies both under a common GEOID schema
- Flow/OD data preparation for vector-based mapping is typically hand-coded per project – `flow2vec` generalizes this step
- Color palette selection for choropleth maps is rarely tested for perceptual spread – `viewPaletteSpread` gives users a diagnostic tool
- Department-specific teaching tool – fully customized to fit UTD's spatial analysis courses

III. Study Area and Data Sources

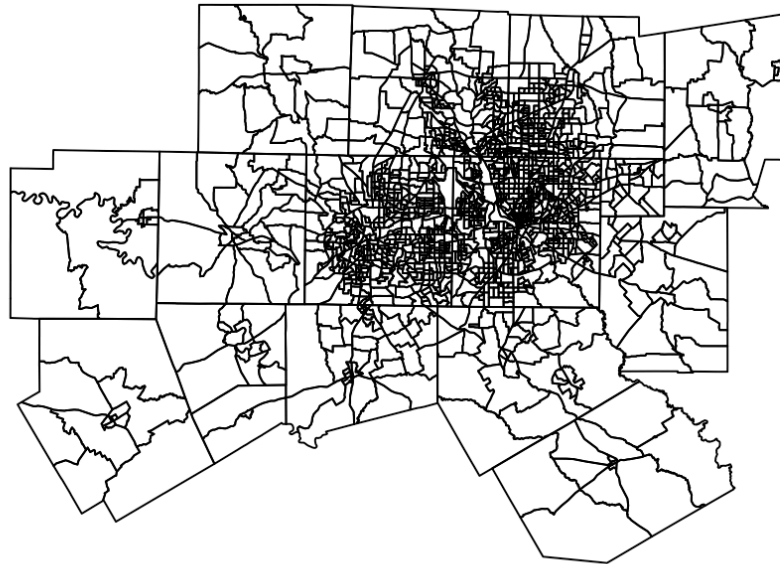
Study Area

- **Primary region:** Dallas County, with buffer data for all North Central Texas Council of Governments (NCTCOG) counties
 - Tract- and blockgroup-level detail
- **Secondary region:** All 254 Texas counties (statewide generalization)
- **For migration vignette** Italy migration data, included for cross-national migration comparison exercises

```
1 library(TexMix)
```

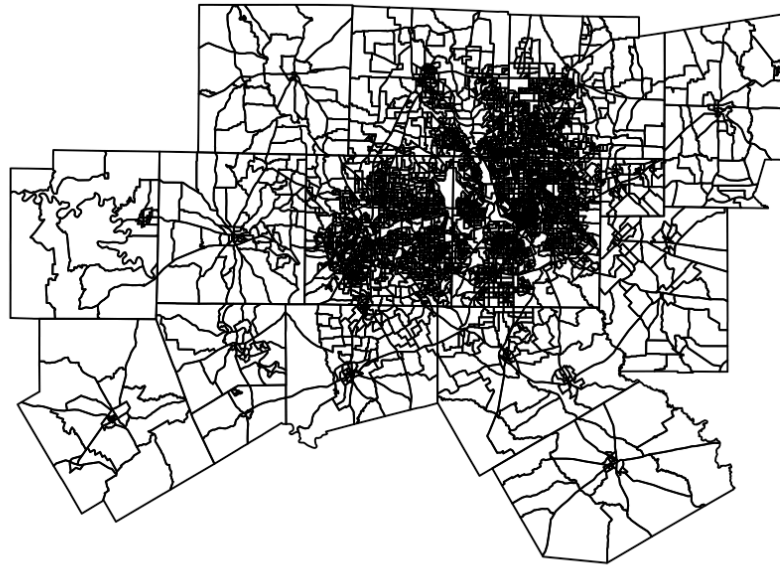
DFW Region – Census Tracts

```
1 data("DFW_tractShp")  
2 plot(DFW_tractShp)
```



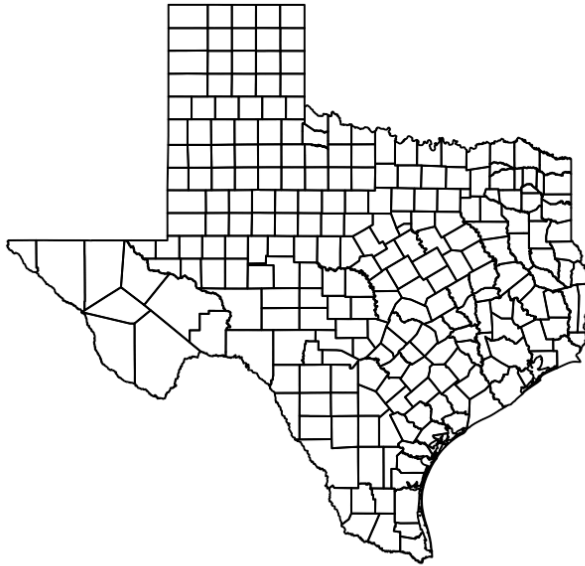
DFW Region – Block Groups

```
1 data("DFW_bgShp")  
2 plot(DFW_bgShp)
```



Texas Counties

```
1 data("TX_countyShp")  
2 plot(TX_countyShp)
```



Italy provinces

```
1 data("ItalyShp")  
2 plot(ItalyShp)
```



New Data – NCTCOG Tracts and Blockgroups

- 2024 ACS data for census tracts and blockgroups within NCTCOG counties
- New factor variables added to support classification and grouping:
 - School district
 - Municipality
 - Congestion level
- These factors enable direct filtering, faceting, and choropleth mapping without additional joins

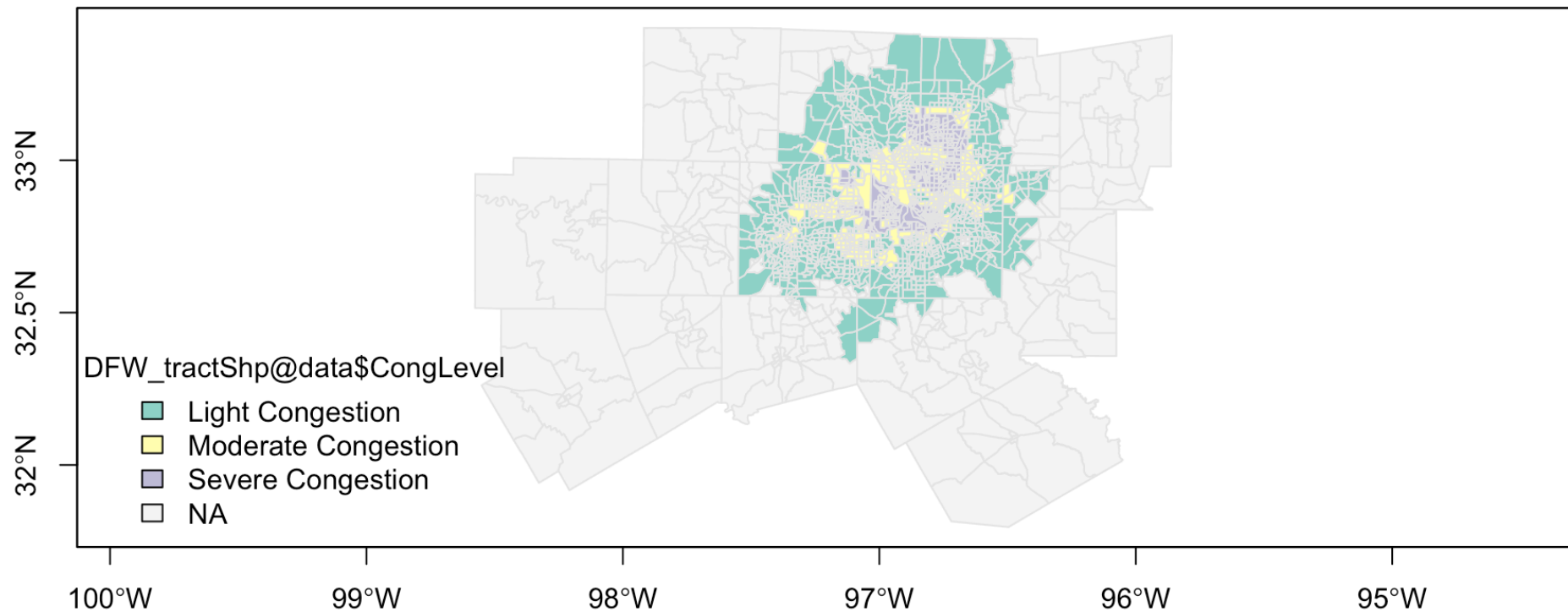
Census data

```
1 colnames(DFW_tractShp@data)
```

[1]	"GEOID"	"NAME"	"LANDAREA"	"WATERAREA"
[5]	"NIGHTPOP"	"POPDEN"	"MALE"	"FEMALE"
[9]	"MEDAGE"	"PCTWHITE"	"PCTBLACK"	"PCTASIAN"
[13]	"PCTHISPAN"	"PCTMINOR"	"PCTNOHIGH"	"PCTUNIVDEG"
[17]	"PCTPUB2WRK"	"HHMEDINC"	"MEDFAMINC"	"PERCAPINC"
[21]	"PCTPOPP0V"	"PCTFAMPOV"	"PCTUNEMP"	"PCTHUVAC"
[25]	"MEDVALHOME"	"PCTNOVEH"	"PCTNOHINS"	"PCTB2010"
[29]	"PCTB2000"	"PCTB1990"	"PCTB1980"	"PCTB1970"
[33]	"PCTB1960"	"PCTB1950"	"PCTB1940"	"PCTBPRE"
[37]	"SchoolDistrict"	"Municipality"	"CongLevel"	

New Factor Data Example

```
1 mapColorQual(DFW_tractShp@data$CongLevel, DFW_tractShp)
```



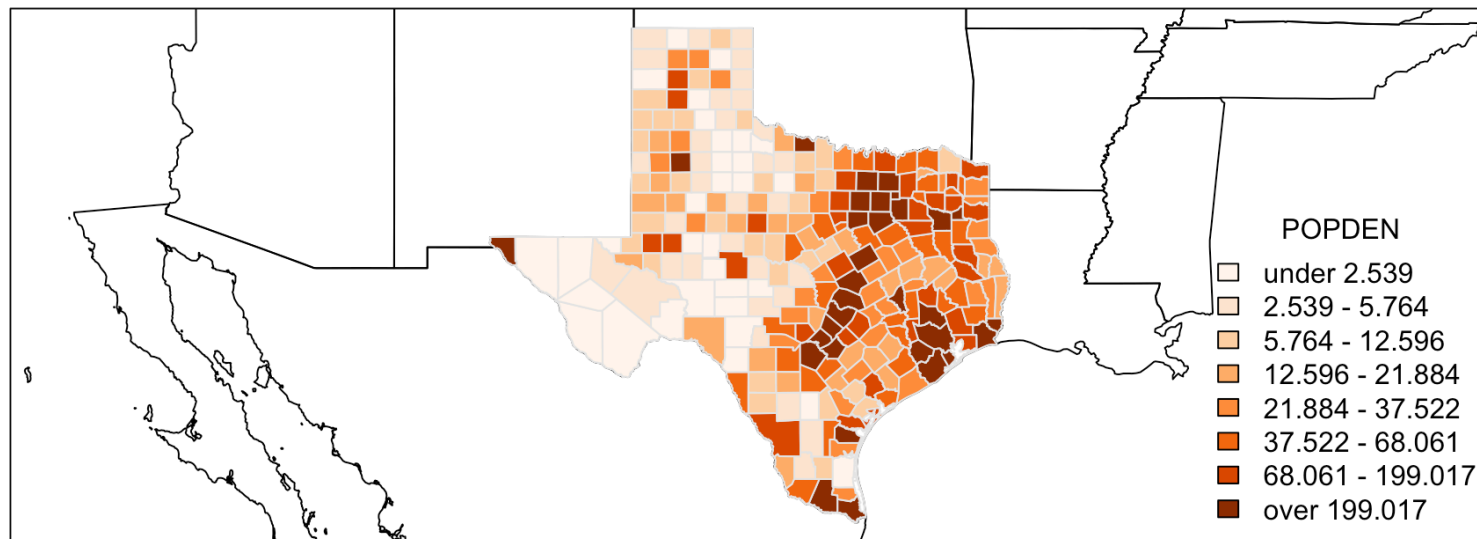
New Data – Statewide and Spatial Layers

- 2024 ACS data extended to **all Texas counties**, enabling state-level context alongside the NCTCOG focus
- **Expanded highway and water layers** for richer basemap cartography and spatial reference
- **Texas neighbors sdf**: a spatial data frame of bordering states, for cleaner basemaps on any Texas-extent map
- **Italy migration data and basemap files**: adds an international migration dataset for comparative coursework and research

Example – Texas neighbors

```
1 data("TX_neighShp")
2 extent <- bbox(TX_countyShp)
3 plot(TX_neighShp, xlim=extent[1,], ylim=extent[2,])
4 mapColorRamp(TX_countyShp, "POPDEN", add.to.map = T, legend.pos="bottomright")
```

Sequential Choropleth Map



Data Design Principle: Universal GEOID

- Every new dataset in this release includes a **GEOID** field
- Rationale:
 - Consistent join key across tracts, blockgroups, and counties
 - Simplifies adding future variables without re-engineering joins
 - Enables long-term maintainability as ACS vintages update annually

IV. Analysis / Methodology

New Function: `flow2vec`

- Converts origin-destination / migration flow tables into vector-ready formats for mapping and analysis
- Formerly implemented as `prepIJDf` in an in-class script; renamed and substantially rebuilt
- **Performance improvement:** parallel calculations across columns are now vectorized, rather than looped, reducing processing time on large flow tables

Example: flow2vec

```
1 data("ItalyShp")
2 data("Italy_migration")
3
4 df <- ItalyShp@data[,c("TOTPOP94","TOTFERTRAT")]
5 mij <- as.matrix(Italy_migration[,2:96])
6 vec.migration <- flow2vec(df, mij=mij)
7 head(vec.migration)
```

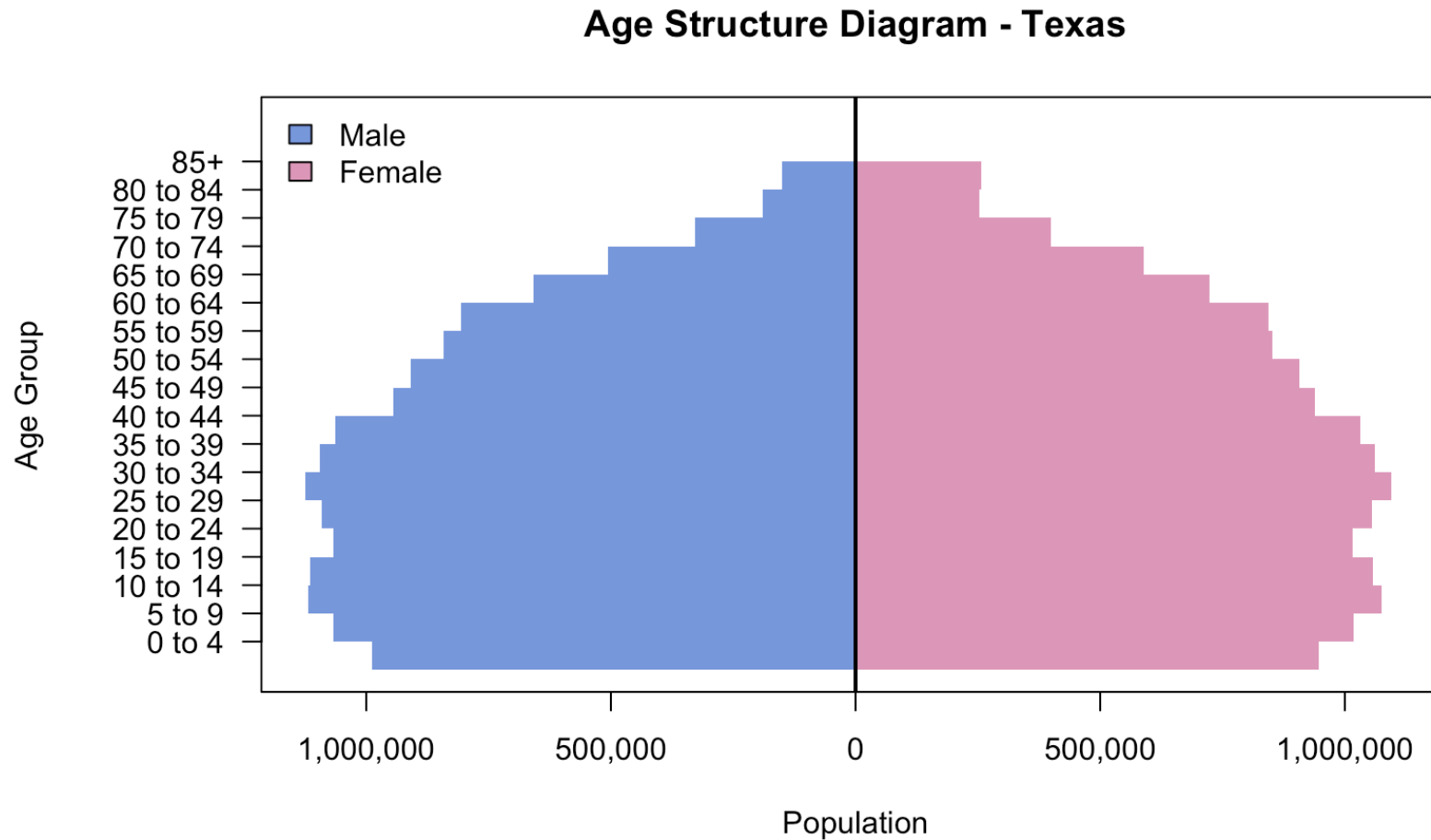
	ID.i	ID.j	ID.ij	mij	dij	TOTPOP94.i	TOTPOP94.j	TOTPOP94.ijL	TOTFERTRAT.i
1	1	1	101	0	NA	292298	292298	0.0000000	1.29
2	2	1	201	16899	NA	387305	292298	0.2814387	1.19
3	3	1	301	1571	NA	239829	292298	-0.1978477	1.29
4	4	1	401	1938	NA	699407	292298	0.8724590	1.53
5	5	1	501	1160	NA	1554178	292298	1.6709282	1.46
6	6	1	601	281	NA	402118	292298	0.3189717	1.38
	TOTFERTRAT.j		TOTFERTRAT.ijL						
1		1.29		0.00000000					
2		1.29		-0.08068891					
3		1.29		0.00000000					
4		1.29		0.17062552					
5		1.29		0.12379422					
6		1.29		0.06744128					

New Function: **ageStructure**

- Generates standardized age-structure diagram (population pyramid) from ACS age variables
- Reduces repetitive recoding of age bins across projects
- Designed to work directly with the new NCTCOG and statewide ACS extracts

Example: ageStructure

```
1 ageStructure(TX_countyShp, name="Texas")
```



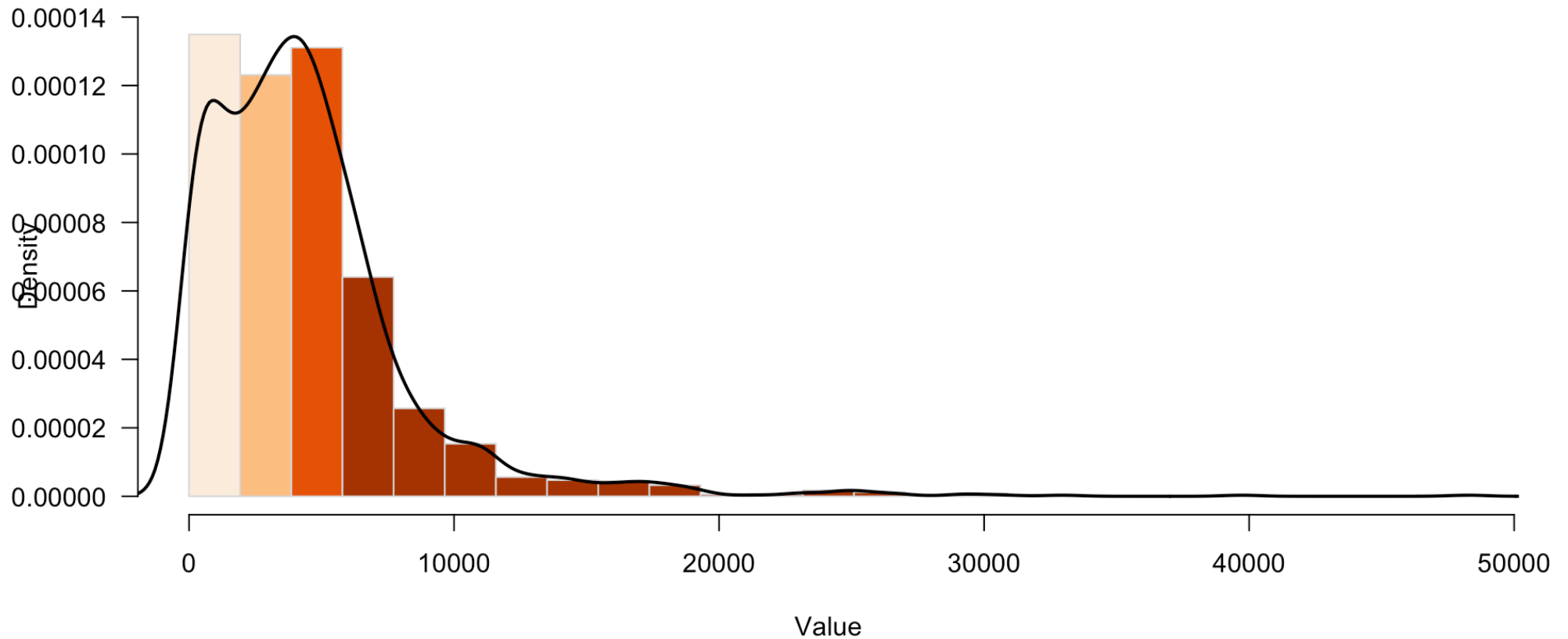
New Function: `viewPaletteSpread`

- Diagnostic tool for evaluating color palettes before mapping
- Helps users visually check perceptual spacing/contrast of a palette across classes
- Complements the updated `mapColorRamp` function by letting users see how observations are distributed within the palette

Example: viewPaletteSpread

```
1 viewPaletteSpread(ramp)
```

Distribution

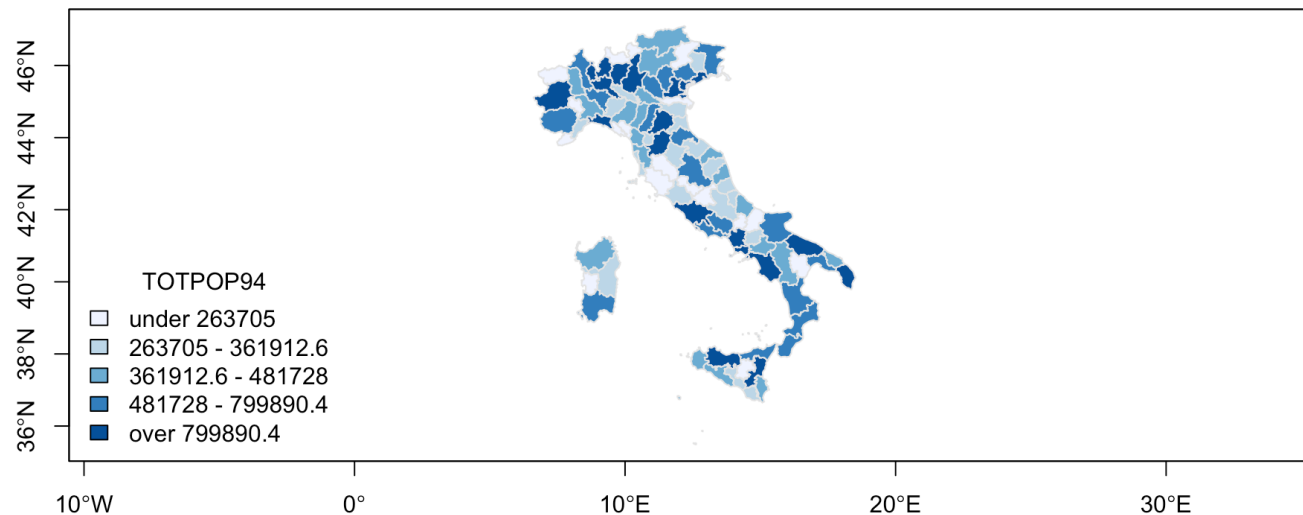


Updated Function: `mapColorRamp`

- Refined to provide more palette options and user-control over the plot

```
1 mapColorRamp(ItalyShp, 'TOTPOP94', breaks=5, palette = "Blues")
```

Sequential Choropleth Map



Methodological / Technical Updates

- **Vectorization:** `flow2vec` parallelizes calculations across columns instead of row-wise iteration, improving performance on large datasets
- **GEOID standardization:** all new datasets carry GEOID for easy joins and future variable additions
- **Testing framework:** new functions (`flow2vec`, `ageStructure`, `viewPaletteSpread`) are now covered by automated tests, supporting long-term reliability and CRAN submission readiness

V. Results and Discussion

What This Release Enables

- Analysts can now move between **regional (NCTCOG)** and **statewide (Texas)** scales using the same GEOID-based join logic
- Migration/commuting flow work is faster and less error-prone with `flow2vec`'s vectorized approach
- Age-structure analysis and thematic color mapping are now first-class, tested functions rather than ad hoc scripts
- The new Texas neighbors sdf and expanded highway/water layers improve basemap quality for any Texas-focused map

Known Considerations

- New factor variables (school district, municipality, congestion level) can shift year to year
- Test coverage should be expanded to include existing functions, not just the three new functions
- old data needs to be phased out of the package

VI. Conclusions and Future Study

Conclusions

- This release meaningfully expands TexMix's data footprint (NCTCOG + statewide Texas + Italy) and analytical toolkit (`flow2vec`, `ageStructure`, `viewPaletteSpread`)
- Technical investments – vectorization, universal GEOID, and a testing framework – position the package for more sustainable, reliable growth
- TexMix is closer to being a robust, general-purpose teaching and research tool for Texas spatial demography

Next Steps

- **Submit final package to CRAN**
- **Set up an official GitHub repository**
- **Build compiled tarballs for all supported systems**
- **Gather feedback from students using the package in coursework**

Future Study

- Expand the testing framework to cover all exported functions, including updated ones
- Annual ACS refresh workflow, leveraging the new GEOID standardization
- Long term: will need to migrate to `sf` package once `sp` is decommissioned

VII. References / Bibliography

References

- U.S. Census Bureau. *American Community Survey (ACS) 2024, 5-Year Estimates*.
- North Central Texas Council of Governments (NCTCOG). *Regional Data and GIS Resources*.
- Pebesma, E. (2018). *Simple Features for R: Standardized Support for Spatial Vector Data*. The R Journal.
- Walker, K. (2023). *Analyzing US Census Data: Methods, Maps, and Models in R*.
- TexMix package documentation and prior release notes (internal).

Thank You

- Questions and feedback welcome
- Package repository, documentation, and CRAN listing forthcoming per the next-steps roadmap